INCH-POUND
MIL-PRF-22885/83F
31 January 2026
SUPERSEDING
MIL-PRF-22885/83E
w/Amendment 1
14 March 2016

PERFORMANCE SPECIFICATION SHEET

# SWITCH, PUSH BUTTON, ILLUMINATED, (2.330 LONG X .750 SQUARE)

INACTIVE FOR NEW DESIGN
AFTER 27 JUNE 1990

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the indicator lights described herein shall consist of this specification sheet and MIL-PRF-22885.

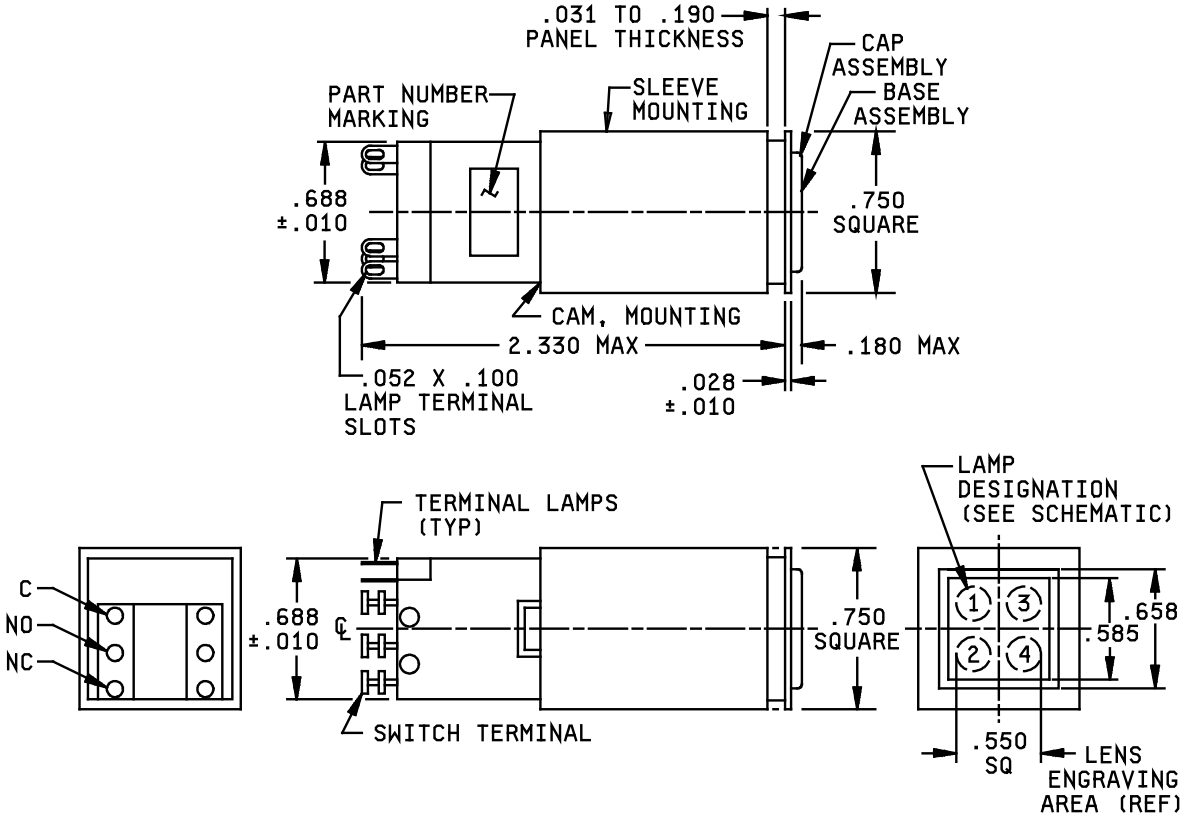
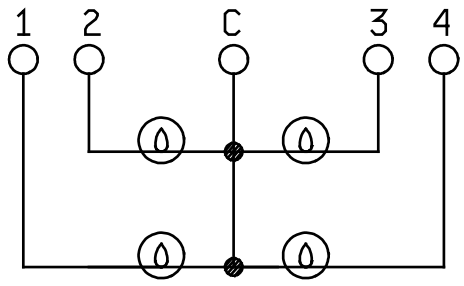
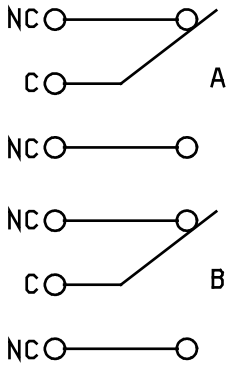



FIGURE 1. Design and construction.

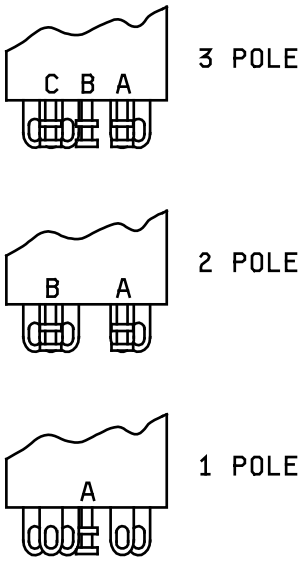

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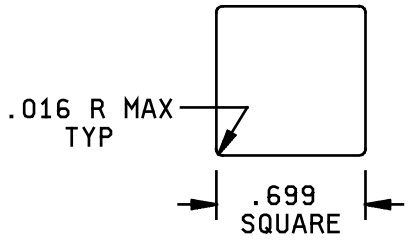



SCHEMATIC DIAGRAM 2PDT SHOWN




BASIC SWITCH IDENTIFICATION



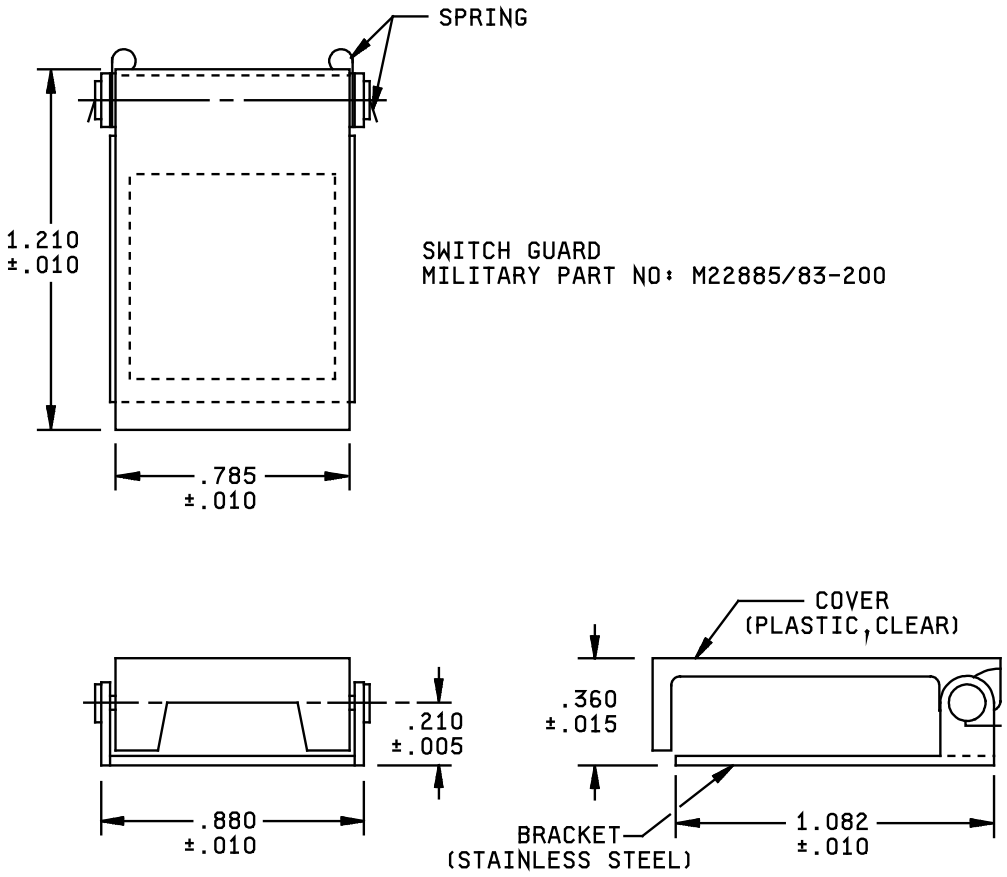

RECOMMENDED MOUNTING PANEL CUTOUT
MOMENTARY OR ALTERNATIVE ACTION

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005.
3. All terminals shall be insulated from the case.
4. All terminals and basic switches shall be permanently identified as shown.
5. Basic switch shall be MIL-PRF-8805/4 switch listed on QPL-8805.
6. Cap assemblies shall be held captive to base assemblies by retaining element to prevent accidental interchange.
7. Exact shape of switch optional provided dimensions specified are not exceeded.

FIGURE 1. Design and construction - Continued.



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NOTES:
1. Dimensions are in inches.
2. The switch guard prevents movement of the actuating means when the guard is in a closed position. The guard shall be maintained in the closed position by a spring mechanism.

FIGURE 2. Switch guard.



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REQUIREMENTS:

Design and construction: See figure 1.

Complete switch shall consist of:
1 - Base assembly - includes MIL-PRF-8805/4 basic switches and mounting sleeve.
1 - Cap assembly - includes necessary color filters and lenses.
4 - Lamps - T 1 3/4 midget flange base, incandescent not included, order separately.

Enclosure design: 1 (unsealed)

Temperature characteristic: 1 (-55°C to 85°C).

Vibration grade: 2 (10-500 Hz, 10 g peak).

Color and luminance: See table I.

Display: Translucent white, which when illuminated shows the required color (type W).

Weight: .080 ± .008 pound.

Operating characteristics:

Actuating force: 1 to 4 pounds.

Pretravel: 0.225 inch maximum.

Cap assembly extraction force: 2.0 ± 1.0 pounds.

Shock (specified pulse): MIL-STD 202-213, test condition H, except 40g.

Electrical ratings: See table II.

Switch guard: See figure 2.

Legends: See table III.

EMI/RFI shielding: Cap-lampholder assembly to have screen and contact for RFI shielding. Shielding efficiency test is applicable. The test shall consist of electric field and plane wave measurements performed at the following frequencies. Attenuation shall not be less than specified below.

| Electric field | Minimum attenuation | Plane wave | Minimum attenuation |
|---|---|---|---|
| 15 kHz | 60 dB | 100 MHz | 47 dB |
| 150 kHz | 60 dB | 400 MHz | 39 dB |
| 500 kHz | 60 dB | 1 GHz | 33 dB |
| 1 MHz | 60 dB | 2.5 GHz | 28 dB |
| 10 MHz | 60 dB | 10 GHz | 20 dB |



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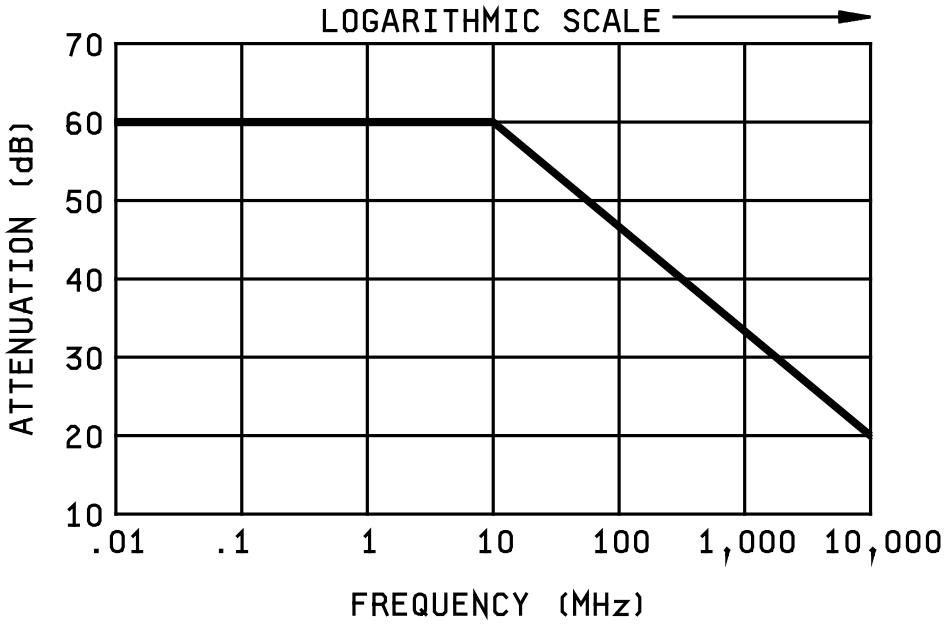



TABLE I. Color and luminance.

| Chromaticity coordinates 2360°K | | | Luminance minimum Light output (foot lamberts) |
|---|---|---|---|
| Colors | X | Y | |
| Red | .695<br>.703<br>.655<br>.660 | .285<br>SL 1/<br>.325<br>SL 1/ | 60 |
| Green | .150<br>.150<br>.300<br>.300 | .450<br>.580<br>.580<br>.450 | 50 |
| Yellow | .562<br>.570<br>.596<br>.605 | .415<br>SL 1/<br>.382<br>SL 1/ | 250 |
| Blue | .200<br>.120<br>.085<br>.175 | .260<br>.222<br>.280<br>.315 | 10 |
| White | .445 Ref. | .425 Ref. | 175 |

1/ SL - Spectrum locus.



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TABLE II. Electrical ratings.

| Load | Sea level | | 50,000 feet |
|---|---|---|---|
| | 28 V dc amperes | 115 V ac (amperes) | 28 V dc amperes |
| Resistive | 7.0 | 7.0 | 4.0 |
| Inductive | 4.0 | 7.0 | 2.5 |
| Lamp | 2.5 | 2.0 | 2.5 |

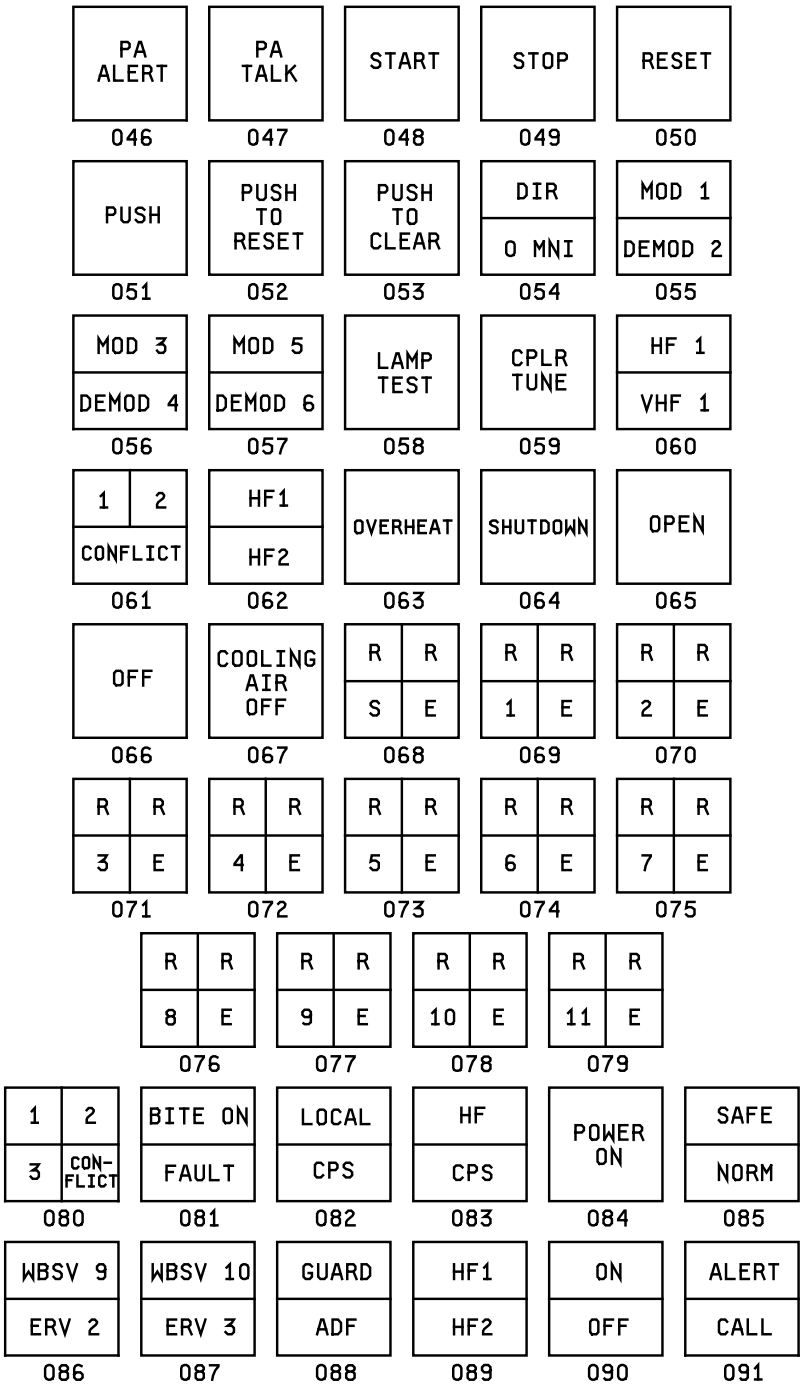
TABLE III. Legends.

| | | | | |
|---|---|---|---|---|
| CLR 1<br>001 | CLR 2<br>002 | CLR 3<br>003 | CLR 4<br>004 | CLR 5<br>005 |
| CLR 6<br>006 | CLR 7<br>007 | CLR 8<br>008 | CLR 9<br>009 | CLR 10<br>010 |
| CLR 11<br>011 | CLR 12<br>012 | CLR 13<br>013 | CLR 14<br>014 | CLR 15<br>015 |
| CLR 16<br>016 | CLR 17<br>017 | CLR 18<br>018 | SEC 1<br>019 | SEC 2<br>020 |
| SEC 3<br>021 | SEC 4<br>022 | SEC 5<br>023 | SEC 6<br>024 | SEC 7<br>025 |
| SEC 8<br>026 | SEC 9<br>027 | SEC 10<br>028 | SEC 11<br>029 | SEC 12<br>030 |
| SEC 13<br>031 | DA<br>032 | MAIN ICS<br>033 | TALK / MON<br>034 | INTPH 1<br>035 |
| INTPH 2<br>036 | SVCE ICS SNLG<br>037 | SVCE ICS<br>038 | CLR / SEC<br>039 | DA1<br>SEC \| CLR<br>040 |
| DA2<br>SEC \| CLR<br>041 | DA3<br>SEC \| CLR<br>042 | DA4<br>SEC \| CLR<br>043 | TEST<br>SEC \| CLR<br>044 | INTPH ALL CALL<br>045 |



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TABLE III. Legends. - Continued.




NOTE: Character height and type shall be 1/8 inch Gorton extra condensed Gothic. Unless otherwise specified in table VII, legends shall be opaque black letters visible at all times on a white background which when illuminated shows the required color.



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PIN: M22885/83- (dash numbers 07 through 97 from table VII). Dash numbers 01 through 06 and any switches without legends shall be assigned as follows:

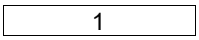
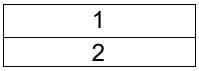
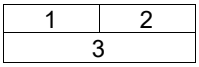
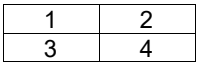
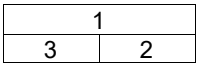
| M22885/83- | F | 5YGR |
|---|---|---|
| Specification Sheet number | Base assembly Code (see table IV) | Cap assembly code (see table V) |
| Example - | 3PDT alternate action | 3-way split lens top illuminates yellow, bottom right illuminates green bottom left illuminates red |

TABLE IV. Base assembly code (circuit and operation).

| Code | Circuit | Operation |
|---|---|---|
| A | 1PDT | Momentary |
| B | 1PDT | Alternate |
| C | 2PDT | Momentary |
| D | 2PDT | Alternate |
| E | 3PDT | Momentary |
| F | 3PDT | Alternate |
| G | Indicator | ---- |
| H 1/ | None | ---- |

1/ Use code H to specify cap assembly portion of switch only.

TABLE V. Cap assembly code (lens style and color).

| Code 1/ | Lens style 2/ |
|---|---|
| 1X |  |
| 2XX |  |
| 3XXX |  |
| 4XXXX |  |
| 5XXX |  |
| 6 3/ | None |

1/ Substitute appropriate color symbol for each "X" in proper sequence.
2/ Numbers indicate color symbol sequence for PIN.
3/ Use code 6 to specify base assembly portion of switch only.



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Qualification inspection: All applicants for qualification approval shall demonstrate that each of their items conforms to all the requirements specified in the applicable documents singularly and in combination with all other previously qualified items, regardless of manufacturer.

Group submission:

a. When the basic sensitive switches are qualified to MIL-PRF-8805, switches shall be tested in accordance with table VI.

b. When the basic sensitive switches are not qualified to MIL-PRF-8805, the switch type for which qualification is sought, shall be submitted tot he complete test in accordance with the qualification inspection table of MIL-PRF-22885.

c. The following applies when the momentary and the alternate action switches are submitted for qualification at the same time.

TABLE VI. Extent of qualification.

| Switch | Test of MIL-PRF-22885 qualification inspection table | Extent of approval |
|---|---|---|
| M22885/83-F1W (all sample units) | Groups I, II, II, VI, VII, and VIII 1/ 2/ | All |
| M22885/83-E1W (2 sample units) | Group I and vibration and shock | |

1/ Cap assemblies containing green, red, blue and yellow filters shall be utilized with M22885/83-FG base assembly for group VIII tests.
2/ Electrical endurance, dc resistive load only.

Group B inspection: Not applicable.

TABLE VII. Dash number and characteristics.

| Dash number 1/ | Legends (see table III code) | Circuit characteristics | Operation | illuminated color |
|---|---|---|---|---|
| 07 | 001 | 3PDT | Alternate | Green |
| 08 | 002 | 3PDT | Alternate | " |
| 09 | 003 | 3PDT | Alternate | " |
| 10 | 004 | 3PDT | Alternate | " |
| 11 | 005 | 3PDT | Alternate | " |
| 12 | 006 | 3PDT | Alternate | " |
| 13 | 007 | 3PDT | Alternate | " |
| 14 | 008 | 3PDT | Alternate | " |
| 15 | 009 | 3PDT | Alternate | " |
| 16 | 010 | 3PDT | Alternate | " |
| 17 | 011 | 3PDT | Alternate | " |
| 18 | 012 | 3PDT | Alternate | " |
| 19 | 013 | 3PDT | Alternate | " |

See footnotes at end of table.



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| Dash number 1/ | Legends (see table III code) | Circuit characteristics | Operation | illuminated color |
|---|---|---|---|---|
| 20 | 014 | 3PDT | Alternate | Green |
| 21 | 015 | 3PDT | Alternate | " |
| 22 | 016 | 3PDT | Alternate | " |
| 23 | 017 | 3PDT | Alternate | " |
| 24 | 018 | 3PDT | Alternate | " |
| 25 | 019 | 3PDT | Alternate | Red |
| 26 | 020 | 3PDT | Alternate | " |
| 27 | 021 | 3PDT | Alternate | " |
| 28 | 022 | 3PDT | Alternate | " |
| 29 | 023 | 3PDT | Alternate | " |
| 30 | 024 | 3PDT | Alternate | " |
| 31 | 025 | 3PDT | Alternate | " |
| 32 | 026 | 3PDT | Alternate | " |
| 33 | 027 | 3PDT | Alternate | " |
| 34 | 028 | 3PDT | Alternate | " |
| 35 | 029 | 3PDT | Alternate | " |
| 36 | 030 | 3PDT | Alternate | " |
| 37 | 031 | 3PDT | Alternate | " |
| 38 | 032 | 3PDT | Alternate | White |
| 39 | 033 | 3PDT | Alternate | " |
| 40 | 034 | 3PDT | Alternate | " |
| 41 | 035 | 3PDT | Alternate | " |
| 42 | 036 | 3PDT | Alternate | " |
| 43 | 037 | 3PDT | Alternate | " |
| 44 | 038 | 3PDT | Alternate | " |
| 45 | 039 | 3PDT | Alternate | 2/ |
| 46 | 040 | 3PDT | Alternate | " |
| 47 | 041 | 3PDT | Alternate | " |
| 48 | 042 | 3PDT | Alternate | " |
| 49 | 043 | 3PDT | Alternate | " |
| 50 | 044 | 3PDT | Alternate | " |
| 51 | 045 | 3PDT | Alternate | White |
| 52 | 046 | 3PDT | Momentary | " |
| 53 | 047 | 3PDT | Momentary | " |
| 54 | 048 | 2PDT | Momentary | Blue 3/ |
| 55 4/ | 049 | 2PDT | Momentary | " |
| 56 | 050 | 2PDT | Momentary | " |
| 57 4/ | 051 | 1PDT | Momentary | White 3/ |
| 59 | 053 | 1PDT | Momentary | " |
| 60 | 052 | 1PDT | Momentary | Yellow |
| 61 | 054 | 1PDT | Alternate | Green |
| 65 | 058 | 3PDT | Momentary | White |
| 66 | 059 | Indicator | --- | " |
| 67 | 060 | Indicator | --- | " |

See footnotes at end of table.



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| Dash number 1/ | Legends (see table III code) | Circuit characteristics | Operation | illuminated color |
|---|---|---|---|---|
| 68 | 061 | Indicator | --- | White |
| 69 5/ | 048 | 2PDT | Momentary | --- |
| 70 5/ | 049 | 2PDT | Momentary | --- |
| 71 | 050 | 2PDT | Momentary | --- |
| 72 5/ | 053 | 1PDT | Momentary | --- |
| 73 6/ | 062 | 1PDT | Momentary | Yellow |
| 78 | 067 | 2PDT | Momentary | Red |
| 79 6/ | 055 | 2PDT | Momentary | Green |
| 80 6/ | 056 | 2PDT | Momentary | Green |
| 81 6/ | 057 | 2PDT | Momentary | Green |
| 82 6/ | 063 | 1PDT | Momentary | Yellow |
| 83 7/ | 064 | 1PDT | Momentary | Red |
| 84 7/ | 065 | Indicator | --- | White |
| 85 7/ | 066 | Indicator | --- | Yellow |
| 86 | 068 | 2PDT | Alternate | 8/ |
| 87 | 069 | 2PDT | Alternate | 8/ |
| 88 | 070 | 2PDT | Alternate | 8/ |
| 89 | 071 | 2PDT | Alternate | 8/ |
| 90 | 072 | 2PDT | Alternate | 8/ |
| 91 | 073 | 2PDT | Alternate | 8/ |
| 92 | 074 | 2PDT | Alternate | 8/ |
| 93 | 075 | 2PDT | Alternate | 8/ |
| 94 | 076 | 2PDT | Alternate | 8/ |
| 95 | 077 | 2PDT | Alternate | 8/ |
| 96 | 078 | 2PDT | Alternate | 8/ |
| 97 | 079 | 2PDT | Alternate | 8/ |
| 98 | 080 | Indicator | --- | White |
| 99 | 081 | 2PDT | Alternate | 9/ |
| 100 | 082 | Indicator | --- | White |
| 101 | 083 | Indicator | --- | White |
| 102 | 084 | 2PDT | Alternate | Green |
| 103 | 085 | 2PDT | Alternate | 10/ |
| 104 6/ | 086 | 1PDT | Alternate | Green |
| 105 6/ | 087 | 1PDT | Alternate | Green |
| 106 13/ | 088 | 2PDT | Alternate | Green |
| 107 13/ | 089 | 2PDT | Alternate | Green |
| 108 | 090 | 2PDT | Alternate | 11/ |
| 109 | 091 | 2PDT | Momentary | 12/ |

See footnotes on next page.



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1/ Dash numbers 01 through 06 (switches without legends) from MIL-PRF-22885/83B have been replaced by A1W through F1W, respectively.
2/ Areas marked CLR will illuminate green when activated.
Areas marked SEC will illuminate red when activated.
All other areas will illuminate with when activated.
3/ Visible letters on a black opaque background. Letters show color when illuminated.
4/ These dash numbers are for switches with switch guards.
5/ These dash numbers are for nonilluminated switches (lamp terminals are not supplied).
6/ Terminal C is common for lamps 2 and 3 and terminal D is common for lamps 1 and 4.
7/ Luminance value is 15 ±5 foot lamberts.
8/ Areas marked R will illuminate green when activated. Areas marked E will illuminate yellow when activated. All other areas will illuminate white when activated.
9/ Area marked 'BITE ON' illuminates white when activated. Area marked 'FAULT' illuminates red when activated.
10/ Hidden legend on black opaque background. Letters show color when illuminated. 'SAFE' illuminates yellow when activated. 'NORM' illuminates green when activated.
11/ Area marked 'ON' illuminates green when activated. Area marked 'OFF' illuminates white when activated.
12/ Area marked 'ALERT' illuminates yellow when activated. Area marked 'CALL' illuminates white when activated.
13/ Shall be 150 foot lamberts luminance minimum and shall be readable in 10,000 foot candles ambient. Hidden legend on black opaque background.

Referenced documents. In addition to MIL-PRF-22885, this document references the following:
MIL-PRF-8805/4 MIL-STD-202-213 MIL-PRF-8805 QPL-8805

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